



Reg. No:

--	--	--	--	--	--	--	--	--	--	--

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS259 - FOUNDATIONS OF BLOCK CHAIN TECHNOLOGY

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1 When is the following a key challenge addressed by the Byzantine Agreement in blockchain?**
 - a) Data compression
 - b) Transaction speed
 - c) Fault tolerance
 - d) Smart contract deployment
2. **CO1 The AAP (Asynchronous Agreement Protocol) is used to _____.**
 - a) Encrypt blockchain data
 - b) Achieve consensus without assuming synchronized clocks
 - c) Reduce transaction fees
 - d) Increase hash power
3. **CO2 Which of the following is used to verify the authenticity of transactions in cryptocurrencies?**
 - a) Signature schemes
 - b) Encryption schemes
 - c) Blockchain
 - d) Public Key Infrastructure (PKI)
4. **CO2 Elliptic Curve Cryptography is primarily used for which purpose in cryptocurrency?**
 - a) Transaction verification
 - b) Key generation
 - c) Data encryption
 - d) Blockchain hashing
5. **CO3 Name the role of the Ethereum Virtual Machine.**
 - a) Manages node identity
 - b) Executes smart contracts
 - c) Stores user data
 - d) Mines ether

PART - B

(5x 3 = 15 Marks)

6. **CO1** Discuss the formal treatment of 'consistency,' 'liveness,' and 'fairness' in blockchain protocols. Why are these important?
7. **CO1** Illustrate how Nakamoto Consensus achieves trustless consensus in a permission-less blockchain.
8. **CO2** Estimate how the Merkle Tree improves the efficiency of transaction verification in Bitcoin.
9. **CO2** Discuss the difference between a private key and a public key in the context of digital signatures in cryptocurrency.
10. **CO3** Describe a simple scenario in which a reentrancy attack can occur.

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) Define abstract blockchain models. Discuss briefly the GARAY and RLA models. (5)
CO1 ii) Calculate the advantages and disadvantages of PoW in blockchain consensus. (5)
12. **CO1** i) Describe the Proof of Stake mechanism and validator selection. (5)
CO1 ii) Calculate a real-world example of a hybrid model like Decred or Ethereum 2.0. (5)
13. **CO2** i) Discuss how Bitcoin provides pseudonymity and how user identities are concealed in transactions. (5)
CO2 ii) Analyze how the difficulty adjustment ensures the stability of Bitcoin's block time and its impact on the security of the network. (5)
14. **CO2** i) Explain the concept of a hard fork and a soft fork in the Bitcoin blockchain. (5)
CO2 ii) Discuss the role of miners in verifying transactions and ensuring that double spending is detected and prevented. (5)

15. **CO3** i) Differentiate SNARKs from interactive zero-knowledge proofs. (5)
CO3 ii) Show the use of Corda in enterprise-level applications such as banking or insurance. (5)
16. **CO3** i) Discuss the concept of pairing-based cryptography and its mathematical basis. (5)
CO3 ii) Compare Hyperledger Fabric and Sawtooth in terms of consensus and architecture. (5)
17. **CO1** i) Define nameless and peer-to-peer. Predict the importance of identity abstraction in such decentralized networks. (5)
CO3 ii) Calculate the benefits of using blockchain in land registration or toll collection systems. (5)
